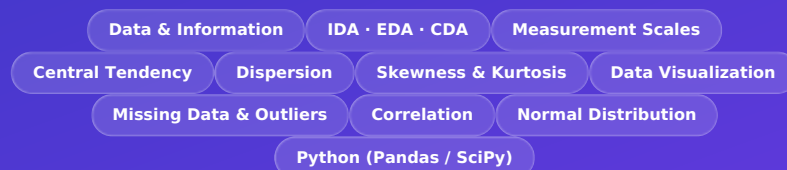


IT 462 • COMPLETE STUDY NOTES • EXPANDED EDITION

Exploratory Data Analysis

From Raw Data to Meaningful Decisions — Full Theory,
Formulas, External References, Worked Examples &
Practice Q&A



LED FROM CLASSROOM LECTURES + STANDARD EDA REFERENCES • SEMESTER EXAM EDITION

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COURSE PHILOSOPHY

Approach this subject with "**An open MIND, a closed HEART**" — stay analytically open to what the data actually shows, and don't let personal bias/assumption ("heart") influence your interpretation of facts. This is the core mindset every data analyst must practice.

What is Exploratory Data Analysis?

Exploratory Data Analysis (EDA) is an approach to analyzing datasets in order to summarize their main characteristics, often using visual and statistical methods, before applying formal modeling or hypothesis testing. It is the process of "getting to know your data" — understanding its structure, spotting patterns, detecting anomalies, testing assumptions, and checking that the data is fit for the questions you want to answer.

A BIT OF HISTORY

The term "Exploratory Data Analysis" was coined and popularized by the American mathematician/statistician **John W. Tukey** in his landmark 1977 book "*Exploratory Data Analysis*." Tukey argued that statisticians had over-focused on **confirmatory** techniques (hypothesis testing) and under-used simple, flexible, visual techniques to genuinely explore data first. He is also credited with inventing the **box plot (box-and-whisker plot)** — one of the most iconic EDA tools, discussed later in Unit 14.

Objectives of EDA

- Understand the underlying structure and distribution of the dataset
- Detect outliers, errors, and anomalies before they distort a model
- Identify important variables and relationships between them
- Check underlying assumptions of subsequent statistical models (e.g., normality, linearity)
- Develop parsimonious models and formulate hypotheses that can later be tested formally (CDA)
- Provide a basis for further data collection through surveys or experiments if needed

Approaches to EDA

GRAPHICAL EDA

Uses visuals to explore data: histograms, box plots, scatter plots, bar charts, heatmaps (detailed fully in Unit 17). Often the fastest way to spot a pattern the human eye can catch instantly but a summary statistic might hide.

NON-GRAPHICAL EDA

Uses numerical summaries: mean, median, mode, variance, standard deviation, skewness, kurtosis, correlation coefficients, frequency tables — the statistical measures covered across Units 12–19.

EDA can further be classified by the number of variables examined at once:

Type	# Variables	Typical Techniques
Univariate	1	Histogram, box plot, mean/median/mode, frequency table
Bivariate	2	Scatter plot, correlation coefficient, cross-tabulation
Multivariate	3+	Heatmaps, pair plots, PCA, multiple regression

Tools Used in this Course

Python

→ Google Colab

→ NumPy

→ Pandas

Python is the primary programming language, run inside **Google Colab** (a free, cloud-based Jupyter notebook environment provided by Google that requires no local setup and gives free access to compute resources). The core libraries relevant to EDA are:

Library	Purpose
NumPy	Numerical computing — n-dimensional arrays, vectorized mathematical operations, linear algebra
Pandas	Data manipulation — reading/cleaning tabular data via the DataFrame object, filtering, grouping, merging
Matplotlib	Foundational plotting library — the base for most charts in Python
Seaborn	Built on Matplotlib — statistical visualizations (box plots, heatmaps, pair plots) with nicer defaults
SciPy	Scientific computing — statistical tests, correlation coefficients (scipy.stats)

EXAM TIP

Examiners often ask "why Python for EDA?" — Answer: it's open-source, free, has a huge ecosystem of statistical and visualization libraries, integrates smoothly with machine learning workflows (scikit-learn, TensorFlow), and has a large community for support.

EXTERNAL REFERENCE

Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley. — the foundational text that formalized EDA as a discipline.

DEFINITION

Data and **Information** are complementary to each other — **data** provides the raw facts, while **information** gives meaning to those facts after processing and analysis.

Worked Illustration✓ **EXAMPLE (FROM CLASS)**

Raw Data: 17, 41, 30, 99 → these are just numbers with no context.

Information: "Scores with an average (AVG) of 50" — once we process the raw numbers (e.g., compute the mean), they become meaningful, usable information.

The DIKW Hierarchy (extended theory)

A widely-used model in information science extends the Data → Information relationship into four layers, often called the **DIKW pyramid**:

Layer	Meaning	Example
Data	Raw, uninterpreted facts/symbols/numbers	"32°C"
Information	Data given context/structure — answers "who, what, where, when"	"Temperature in Ahmedabad today is 32°C"
Knowledge	Information combined with experience/patterns — answers "how"	"32°C in August is typical monsoon-season weather here"
Wisdom	Applied knowledge used to make good judgments — answers "why/should"	"Because it's humid at 32°C, outdoor events should be scheduled for evening"

EDA operates mainly at the **Data → Information** transition — turning numbers into patterns and summaries a decision-maker can use.

DATA (RAW / UNORGANIZED)

Raw, unorganized, **unstructured** facts collected before any processing. Has no inherent meaning by itself. *Data can also be structured* — i.e., already arranged into rows & columns.

INFORMATION

The processed, organized, meaningful output derived from data. Common forms include:

- Text
- Audio
- Visuals
- Records

Characteristics of Good-Quality Data

Before data can become trustworthy information, it should ideally satisfy these standard data-quality dimensions used across the industry:

Dimension	Meaning
Accuracy	Correctly reflects the real-world value it represents
Completeness	No missing values where data is expected
Consistency	No contradictions across different records/sources
Timeliness	Up to date and available when needed
Validity	Conforms to the defined format/range/rules
Uniqueness	No unwanted duplicate records

Sources of Data

Based on **who collects the data and how directly they are involved**, data sources are classified into three types:

Source Type	Meaning	Involvement
Primary Data	Data collected first-hand by the researcher for a specific purpose	I am involved
Secondary Data	Data collected by someone else for their own purpose, which you then reuse	You are involved
Tertiary Data	Data already compiled/summarized from both primary and secondary sources	Compiled from Primary + Secondary

Common Methods of Collecting Primary Data

Method	Description	Example
--------	-------------	---------

Survey / Questionnaire	A structured set of questions given to respondents	Google Forms customer satisfaction survey
Interview	Direct one-on-one or group conversation	HR interview feedback
Observation	Recording behavior/events as they naturally occur	Counting footfall in a store
Experiment	Controlled test manipulating one variable to observe effect	A/B testing a website design

Common Sources of Secondary Data

- Government publications and census reports (e.g., data.gov.in, Census of India)
- Company/organization internal records and financial reports
- Academic journals, research papers, published studies
- Open datasets — Kaggle, UCI Machine Learning Repository, World Bank Open Data

🔑 MEMORY TRICK

Primary = "I collected it myself." **Secondary** = "Someone else collected it, but I use it directly." **Tertiary** = "It's already a summary/compilation of primary + secondary sources" (e.g., encyclopedias, meta-analyses, literature reviews).

⚠️ ADVANTAGES & DISADVANTAGES

Primary data is more accurate/relevant to your exact question but is time-consuming and expensive to collect. **Secondary data** is quick and cheap to access but may not perfectly fit your research question, could be outdated, or may have unknown quality issues.



Types & Sources of Data

Organization, content-forms & the structured/unstructured spectrum

By Degree of Organization

In real-world data science, data organization is often described as a spectrum rather than just two categories:

Type	Description	Example
Structured	Organized into a fixed rows & columns / relational format, easily searchable	SQL database, Excel spreadsheet
Semi-structured	Has some organizational tags/markers but not a rigid schema	JSON, XML, emails (headers + body)
Unstructured	No predefined format or model	Free text, images, videos, social media posts, audio recordings

INDUSTRY NOTE

Estimates commonly cited in data-industry literature suggest that roughly **80% of enterprise data is unstructured** (text, images, video, etc.), which is why techniques like NLP and computer vision have become essential complements to traditional structured-data EDA.

By Content Form (what "Information" typically looks like)

Form	Description / Example
Text	Written words — documents, articles, messages, reviews
Audio	Sound recordings — voice, music, calls
Visuals	Images, charts, videos
Records	Structured entries — database rows, forms, logs

By Time Dimension

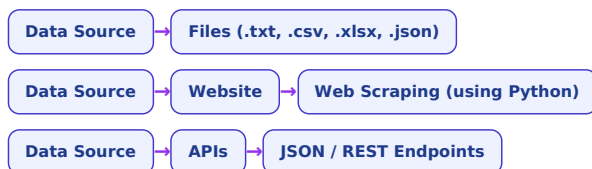
An additional classification relevant to EDA involves how data relates to time:

Type	Description	Example
Cross-sectional data	Observations of multiple subjects at a single point in time	Marks of all students in one exam
Time-series data	Observations of one subject collected repeatedly over time	Daily closing stock price of a company over a year
Panel / Longitudinal data	Combination — multiple subjects observed repeatedly over time	GDP of multiple countries recorded every year for a decade

Data can be stored or represented in different **file formats** depending on its structure and purpose:

Format	Full Meaning	Typical Use
.txt	Plain Text	Unstructured free-form text
.csv	Comma-Separated Values	Simple tabular data — rows & columns, lightweight and universal
.xlsx	Excel Spreadsheet	Tabular data with formatting, formulas, multiple sheets
.json	JavaScript Object Notation	Key-Value pairs — nested/hierarchical data, common in APIs
.html	HyperText Markup Language	Web page content, scraped via web scraping
.xml	eXtensible Markup Language	Tag-based hierarchical data, common in legacy enterprise systems
.parquet	Apache Parquet	Column-oriented binary format, efficient for very large datasets
SQL databases	Structured Query Language tables	Relational data queried directly from a database server

Data Source Categories



Web Scraping — Extended Theory

🔖 CONCEPT NOTE

Web scraping is the technique of programmatically extracting data from HTML web pages, typically automated using Python libraries such as:

`requests` — fetches the raw HTML of a page

`BeautifulSoup` — parses HTML/XML and lets you navigate/search the page's tag structure

`Selenium` — automates a real browser, useful for JavaScript-heavy dynamic pages

`Scrapy` — a full scraping framework for large-scale, multi-page crawling

APIs — Application Programming Interfaces

Many modern data sources (Twitter/X, weather services, financial data providers) expose an **API** that returns data — usually in JSON format — directly, without needing to scrape HTML. APIs are generally the preferred, more reliable and more "polite" way to acquire live data compared to scraping, since they're explicitly designed for programmatic access.

🔗 LOADING DATA IN PANDAS — QUICK REFERENCE

```
pd.read_csv("file.csv") · pd.read_excel("file.xlsx") · pd.read_json("file.json") · pd.read_html("url") · pd.read_sql(query, connection)
```

⚠️ ETHICAL & LEGAL NOTE ON WEB SCRAPING

Always check a website's robots.txt file and Terms of Service before scraping. Scraping copyrighted, private, or rate-limited content without permission can violate a site's terms and, in some jurisdictions, the law.

This is the full journey raw data takes before it drives a business decision:

- ① Data (Raw Facts) ↓
- ② Statistics + Tools (Python, Excel, SQL, R) ↓
- ③ Data Analysis (Summarize data, find patterns, detect unusuality) ↓
- ④ Validation & Data Quality (Validity, Reliability, Accuracy — Error Checking) ↓
- ⑤ Useful Information (Meaningful results / charts / summaries / findings) ↓
- ⑥ Decision / Action (Using insights to make better decisions)

Data Quality Dimensions (as taught)

Before information can be trusted, the underlying data must be checked for:

- **Validity** — does the data conform to expected rules/format?
- **Reliability** — is the data consistent across sources/time?
- **Authenticity** — does it genuinely represent what it claims to?
- **Error-free** — free from data-entry mistakes or impossible values

⚠ WORKED EXAMPLE — SPOTTING BAD DATA

ID	IT462 Score	Age	Gender
2026	10	0	M
2026	50	10	F
2026	90	20	M
2026	-20	10	F
2023	17	...	—

The score **-20** is **invalid** — marks/scores cannot logically be negative. This is a classic *error-checking* scenario: such records fail the "validity" check and must be flagged or corrected before analysis.

Why Data Validation Matters — The "Garbage In, Garbage Out" Principle

This is one of the most widely cited principles in computing and statistics: **if the input data is flawed ("garbage"), no amount of clever analysis or modeling can produce a trustworthy output.** This is precisely why Units 6 and 18 (IDA and Missing Data/Outlier Treatment) exist as formal stages in every analytics workflow — they exist to catch "garbage" before it reaches modeling.

🔗 EXTERNAL REFERENCE

The CRISP-DM (Cross-Industry Standard Process for Data Mining) framework — an industry-standard 6-phase methodology (Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment) — mirrors this same Data→Decision pipeline taught in class, and is worth knowing as a real-world parallel.

6 IDA → EDA → CDA Framework

The three stages of data analysis



1 IDA — INITIAL DATA ACQUISITION

The very first stage where data is gathered and made **usable**. Flow: **Problem → Data Collection → Preprocessing / Cleaning**.

Goal: turn raw, potentially messy data into a usable dataset backed by statistics, tools & packages, while checking **validity, reliability, authenticity, and error-freeness**.

2 EDA — EXPLORATORY DATA ANALYSIS

The core investigative stage, made up of two parallel activities:

Statistical Aspects

- Mean
- Variance
- Mode
- Distribution

Data Processing

- Cleaning
- Missing values
- Outliers
- Transformation

This stage also identifies the **features of data / variables**: Independent (X), Dependent (Y), Numerical, Categorical.

3 CDA — CONFIRMATORY DATA ANALYSIS

The final, hypothesis-driven stage: **test hypotheses and confirm relationships** between variables — includes modelling and evaluation. Unlike EDA (which is open-ended exploration), CDA formally tests specific claims about the data using statistical rigor.

EDA vs CDA — Tukey's Original Distinction (extended theory)

Aspect	EDA	CDA
Purpose	Generate hypotheses, discover patterns	Test pre-specified hypotheses
Approach	Flexible, open-ended, visual	Rigid, rule-based, formal
Tools	Plots, summary statistics	p-values, confidence intervals, significance tests
Analogy	A detective gathering clues	A judge weighing evidence against a specific charge

📌 EXAM TIP — IDA VS EDA VS CDA

IDA = "Is my data ready and clean?" | **EDA** = "What patterns/structure exist in the data?" | **CDA** = "Can I statistically prove this relationship/hypothesis?"

7

Structured Data: Rows & Columns

The anatomy of a dataset

DEFINITION

Structured data is data arranged in a **rows & columns** format, like a spreadsheet or database table.

Component	Also Called	Meaning
Rows	Observations	Tell us who / what we are observing (a single record)
Columns	Variables / Features	Tell us what we are measuring about each observation

QUICK WAY TO REMEMBER

→ **Rows** = observations/records (each row = one entity, e.g., one student)

↓ **Columns** = variables/features/attributes (each column = one measured property, e.g., Age, Gender, Score)

The Pandas DataFrame — Practical Anatomy

In Python's Pandas library, structured data is represented by a **DataFrame** — a 2-dimensional labeled table. Its key components map directly to the theory above:

DataFrame Concept	Corresponds To
df.index	Row labels (observation identifiers)
df.columns	Column labels (variable/feature names)
df.shape	(number of rows, number of columns)
df.dtypes	Data type of each column (int, float, object/string, bool, datetime)
df.head()	Preview the first 5 rows
df.info()	Summary: column names, non-null counts, data types
df.describe()	Summary statistics (mean, std, min, max, quartiles) for numeric columns

FIRST EDA COMMANDS ON ANY NEW DATASET

```
df.shape → how big is it?
df.head() → what does it look like?
df.info() → what types & how much missing data?
df.describe() → what's the numeric summary?
df.isnull().sum() → count missing values per column
```

Variables / Features

Categorical (Qualitative)

Numerical (Quantitative)

CATEGORICAL DATA (QUALITATIVE)Represents **groups or categories**.

- Gender → Male / Female
- Color → Red / Blue
- City → Mumbai / Delhi
- Blood Group → A / B / AB / O

NUMERICAL DATA (QUANTITATIVE)Represents **quantities that can be measured** using numbers.

- Age → 67
- Height → 6.7 m
- Salary → ₹67,000
- Temperature → 67°F
- Weight → 67 kg

Each of these two types is further divided into two **measurement scales** — explored in full detail in Unit 9:

Categorical

Nominal

Ordinal

Numerical

Interval

Ratio

Extended Theory — Discrete vs Continuous Numerical Data

Numerical (quantitative) data is further split, based on what values it can take, into:

Type	Meaning	Example
Discrete	Can only take specific, countable values (often whole numbers); usually arises from counting	Number of children in a family (0, 1, 2, 3...) — cannot have 2.5 children
Continuous	Can take any value within a range (including fractions/decimals); usually arises from measuring	Height (170.2 cm, 170.25 cm...) — infinitely divisible

EXTRA EXAMPLE — POTATO!**Color** of a potato (e.g. brown, yellow) → **Nominal** (no natural order)**Size** of a potato (e.g. small, medium, large) → **Ordinal** (has a natural order)**QUICK SELF-TEST**Is "Number of cars owned" discrete or continuous? → **Discrete** (you can't own 2.5 cars).Is "Distance travelled in km" discrete or continuous? → **Continuous** (can be 5.37 km).

The **4 measurement scales** (remember as **N-O-I-R**) were formally proposed by psychologist **Stanley Smith Stevens** in a famous 1946 paper, "*On the Theory of Scales of Measurement*," and remain the standard framework taught in statistics today. They determine what mathematical operations and statistical tests are valid on a variable.

① NOMINAL CATEGORICAL

Names / labels with **no meaningful order**.

We cannot say "Red > Blue" — there is no such ordering.

Also used for: Names, Indexes, Tags.

More examples: Marital status, Nationality, Type of vehicle, Yes/No responses.

② ORDINAL CATEGORICAL

Has a **meaningful ordering**, but the **difference between ranks isn't measurable**.

e.g. Satisfaction: Poor < Average < Good < Excellent

Education Level: School < Bachelor < Master < PhD

Rank: 3rd < 2nd < 1st

More examples: Movie star ratings (1★–5★), Pain scale (mild/moderate/severe), Military rank.

③ INTERVAL NUMERICAL

Has a **meaningful, equal-interval difference**, but **no true/absolute zero**.

e.g. Temperature (10°C, 20°C, 30°C): The difference is meaningful (10°C → 20°C = 10°C; 20°C → 30°C = 10°C).

But 20°C is *not* twice as hot as 10°C, because 0°C does not mean "no temperature."

More examples: IQ score, Calendar year (0 AD is not "no time"), pH scale.

④ RATIO NUMERICAL

Has **everything Interval has, PLUS a true zero ("0")**.

0 kg = no weight, 0 cm = no height, ₹0 = no income.

Because zero is absolute, ratios are meaningful: **20 kg is twice as much as 10 kg**.

More examples: Distance, Time duration, Number of children, Price.

🕒 SPECIAL CASE — AGE

Age is classified as a **Ratio** scale because it has an absolute zero point (age 0 = birth, meaning "no age has elapsed yet"). Since a true zero exists conceptually, ratio statements are valid: a 20-year-old is twice as old as a 10-year-old.

Master Comparison Table (NOIR)

Property	N (Nominal)	O (Ordinal)	I (Interval)	R (Ratio)
Labelled Data	✓	✓	✓	✓
Inherent Order	✗	✓	✓	✓
Measurable Difference	✗	✗	✓	✓
Absolute / True Zero	✗	✗	✗	✓
Valid Operations	— (none)	— (none)	+, −	+, −, ×, ÷
Valid Central Tendency	Mode only	Mode, Median	Mode, Median, Mean	Mode, Median, Mean

⚠️ COMMON MISTAKE

Students often think Ordinal data can be averaged just because it looks numeric (e.g., ranks 1, 2, 3). But since the "distance" between ranks isn't measurable/equal, arithmetic operations like mean are technically invalid on pure ordinal data — median/mode are preferred.

🔗 EXTERNAL REFERENCE

Stevens, S. S. (1946). "On the Theory of Scales of Measurement." *Science*, 103(2684), 677–680. — the original paper that introduced the Nominal/Ordinal/Interval/Ratio typology used worldwide in statistics courses today.

LIKERT SCALE

An example of an **ordinal scale**, developed by psychologist **Rensis Likert** in 1932 for measuring attitudes in surveys. It is commonly **transformed** to behave like an interval scale by:

- Making a quantifiable equivalence of numbers (assigning numeric values to each response option)
- Making the "difference" between points equal in length

Ordinal → (transform via Likert Scale) → Interval

Typical Likert scale line:



Common Likert formats: **3-point**, **5-point**, **7-point** scales (odd number of points to allow a neutral midpoint). Some survey designs deliberately use an even number of points (a "forced choice" scale) to prevent respondents from defaulting to the neutral middle option.

✓ REAL SURVEY EXAMPLE

"On a scale of 1-5, how satisfied are you with this course?" — this is a single Likert *item*. When several related items are summed/averaged into one score (e.g., a 10-question customer satisfaction index), the result is called a **Likert scale** (as opposed to just one item).

How to Identify the Measurement Scale of a Variable

A simple 4-step decision process to classify any data variable:

Step 1: Can I count / identify groups? → if yes, at least **Nominal**

Step 2: Can I rank / order the values? → if yes, at least **Ordinal**

Step 3: Can I measure the difference between values? → if yes, at least **Interval**

Step 4: Can I find a meaningful ratio (true zero exists)? → if yes, it's **Ratio**

🔑 GOLDEN RULE

First identify the measurement scale of the variable; then determine which operations and statistical tools are appropriate for analyzing it. This single sentence is one of the most exam-relevant lines from the lectures — scale determines the valid statistics.

🔗 EXTERNAL REFERENCE

Likert, R. (1932). "A Technique for the Measurement of Attitudes." *Archives of Psychology*, 22(140), 1-55.

Statistical Tools

Descriptive

Inferential

DESCRIPTIVE STATISTICS

Used to **describe / summarize** the dataset itself, without generalizing beyond it. Key sub-topics: **Central Tendency** (Mean, Median, Mode), **Dispersion** (Range, Variance, SD), **Shape** (Skewness, Kurtosis).

INFERENTIAL STATISTICS

Used to **draw conclusions / make predictions** about a larger **population** based on a **sample** — connects directly to the CDA stage from Unit 6.

Extended Theory — Population vs Sample

Term	Meaning	Notation Example
Population	The entire group you want to study/draw conclusions about	N , μ (mean), σ (SD), σ^2 (variance)
Sample	A subset of the population actually observed/measured	n , $\bar{x}$ (mean), s (SD), s^2 (variance)

Since studying an entire population is often impossible or too costly, we take a representative **sample** and use inferential statistics to estimate population parameters — this is exactly why Unit 13 distinguishes population variance ($+N$) from sample variance ($+(n-1)$).

Core Ideas in Inferential Statistics (introductory theory)

Concept	Meaning
Hypothesis Testing	A formal procedure to test a claim (hypothesis) about a population using sample data
Null Hypothesis (H_0)	The default assumption of "no effect" or "no difference"
Alternative Hypothesis (H_1)	The claim you're testing for — that there IS an effect/difference
p-value	Probability of observing the sample result (or more extreme) if H_0 were true — a small p-value (commonly < 0.05) suggests rejecting H_0
Confidence Interval	A range of values likely to contain the true population parameter, at a stated confidence level (e.g., 95%)

CONNECTING THE DOTS

EDA (Unit 6) mostly produces **descriptive statistics**. CDA (Unit 6) relies on **inferential statistics** — this is precisely why CDA is described as the stage that "tests hypotheses and confirms relationships."

These measures try to find a single "central" value that represents the whole dataset.

Measure	Needs Sorting?	Formula / Rule
Mean ($\bar{x}$)	✗ No	$\bar{x} = \sum x_i / n$ (sum of all values ÷ count)
Median	✓ Yes	If n is odd : median = $x_{((n+1)/2)}$ (the middle value) If n is even : median = $\frac{1}{2} [x_{(n/2)} + x_{(n/2 + 1)}]$ (average of the two middle values)
Mode	✓ Yes	The most frequently occurring value in the dataset

Variants of the Mean

✗ TRIMMED MEAN

Mean computed **after removing the extremes** (x_i for $i \geq 1 \dots n$ at both ends) — reduces the effect of outliers.

🔖 WEIGHTED MEAN

$\bar{x}_w = \sum(w_i x_i) / \sum w_i$ — each value x_i is multiplied by an importance weight w_i before averaging.

Types of Mean

Arithmetic Mean

Geometric Mean

Harmonic Mean

Type	Formula	Best Used For
Arithmetic Mean	$\sum x_i / n$	General-purpose average (default/most common)
Geometric Mean	$\sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$	Rates of growth, multiplicative data (e.g., average annual investment return)
Harmonic Mean	$n / \sum(1/x_i)$	Rates and ratios (e.g., average speed over equal distances)

Which Measure to Use? — Effect of Outliers & Skew

Situation	Best Measure	Why
Symmetric distribution, no outliers	Mean	Uses every data point, most statistically efficient
Skewed distribution / has outliers	Median	Not affected by extreme values (robust)
Categorical / Nominal data	Mode	Only central tendency measure valid for non-ordered categories

⚠️ CLASSIC EXAMPLE — WHY MEAN CAN MISLEAD

Salaries of 5 employees: ₹25,000, ₹27,000, ₹26,000, ₹28,000, ₹5,00,000 (owner).

Mean = $(25,000 + 27,000 + 26,000 + 28,000 + 5,00,000) / 5 = \text{₹1,21,200}$ — very misleading!

Median = **₹27,000** — much more representative of a "typical" employee's salary. This is exactly why real-estate prices, incomes, and similar right-skewed data are usually reported using the **median**, not the mean.

✓ QUICK NUMERICAL EXAMPLE

Data: 2, 3, 5, 6, 8, 10, 12, 15, 18, 20 ($n = 10$, already sorted)

Mean = $(2+3+5+6+8+10+12+15+18+20)/10 = 99/10 = \text{9.9}$

Median ($n = 10$, even) = $\frac{1}{2} (x_5 + x_6) = \frac{1}{2} (8 + 10) = \text{9}$

Mode = No value repeats → this dataset has **no mode** (or is "amodal")

🐍 PYTHON (PANDAS)

```
df['col'].mean() · df['col'].median() · df['col'].mode()
```

13

Measures of Dispersion / Variation

How spread out is the data?

While central tendency tells us the "center," **dispersion** tells us how much the data **spreads out** from that center. Two datasets can have the exact same mean but wildly different spreads — dispersion measures capture that difference.

Measure	Formula	Meaning
Range	$X_H - X_L$	Highest value minus Lowest value — simplest but most sensitive to outliers
Variance (Population, σ^2)	$\sigma^2 = \sum (x_i - \bar{x})^2 / N$	Average squared deviation from the mean, using the entire population (N)
Variance (Sample, s^2)	$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$	Same idea, but divides by (n-1) — Bessel's correction — for an unbiased sample estimate
Standard Deviation	$\sigma = \sqrt{\text{Variance}}$ or $s = \sqrt{\text{Sample Variance}}$	Square root of variance — brings the unit back to the original scale
Coefficient of Variation (CV)	$CV = (s / \bar{x}) \times 100\%$	Relative spread as a % of the mean — lets you compare variability across datasets with different units/scales

🔗 WHY N - 1 FOR SAMPLE VARIANCE?

Dividing by (n - 1) instead of n corrects the natural tendency of a sample to **underestimate** the true population variance — this is called **Bessel's Correction**. Population variance uses N because the entire population is already known.

✓ COEFFICIENT OF VARIATION — WORKED EXAMPLE

Stock A: mean return = 10%, SD = 2% → CV = (2/10)×100 = **20%**

Stock B: mean return = 25%, SD = 4% → CV = (4/25)×100 = **16%**

Even though Stock B has a larger absolute SD (4% vs 2%), it is actually **relatively less volatile** than Stock A once we account for the difference in average returns — this is the power of CV for comparing variability on different scales.

⚠ NOTATION REMINDER

x_H = Maximum score, x_L = Minimum score (from data $x_1 \dots x_n$). Standard deviation is preferred over variance for interpretation because it's in the **same unit** as the original data (variance is in squared units).

🐍 PYTHON (PANDAS)

```
df['col'].std() → sample SD (default, ddof=1) · df['col'].var() → sample variance · df['col'].max() - df['col'].min() → range
```

Percentiles, Quartiles, Quantiles, and Deciles all describe the **position** of a value within a sorted dataset. **Quartile Deviation** is derived from quartiles as another measure of spread.

PERCENTILE FORMULA

For a sorted dataset with **n** values, the percentile position of the *i*-th value x_i is:

$$P(x_i) = \frac{(i - 1)}{(n - 1)} \times 100$$

WORKED EXAMPLE (AS DONE IN CLASS)

Data: 2, 3, 5, 6, 8, 10, 12, 15, 18, 20 → $n = 10$ (positions $i = 1$ to 10)

Q1. What percentile is $x_i = 12$?

12 is at position $i = 7$.

$$P(x_7) = [(7 - 1)/(10 - 1)] \times 100 = (6/9) \times 100 = \mathbf{66.7\%}$$

Q2. Find the 25th percentile (P_{25}).

Rearranging the formula to solve for i : $i - 1 = [P(x_i) \times (n - 1)] / 100$

$$i - 1 = (25 \times 9) / 100 = 225/100 = 2.25$$

$i = \mathbf{3.25}$ (i.e., the 25th percentile lies 25% of the way between the 3rd value (5) and the 4th value (6) in the sorted data)

RELATED TERMS

- Quartiles** — divide data into 4 equal parts (Q1 = 25th percentile, Q2 = 50th = Median, Q3 = 75th percentile)
- Deciles** — divide data into 10 equal parts
- Quantiles** — general term for dividing data into any number of equal parts
- Interquartile Range (IQR)** = $Q3 - Q1$ — the spread of the "middle 50%" of the data
- Quartile Deviation** = $(Q3 - Q1) / 2$ — a robust measure of spread, less affected by outliers than range

Box Plot (Box-and-Whisker Plot) — Extended Theory

Invented by **John Tukey** (see Unit 1), the box plot is one of the most powerful graphical EDA tools because it summarizes 5 key numbers ("the five-number summary") in a single compact picture:

Component	Value
Minimum (excluding outliers)	Smallest value within the "whisker" range
Q1 (Lower Quartile)	25th percentile — bottom edge of the box
Q2 (Median)	50th percentile — line inside the box
Q3 (Upper Quartile)	75th percentile — top edge of the box
Maximum (excluding outliers)	Largest value within the "whisker" range

THE IQR OUTLIER RULE (TUKEY'S FENCES)

A data point is flagged as a potential **outlier** if it falls outside:

$$\text{Lower Fence} = Q1 - 1.5 \times \text{IQR} \quad \text{Upper Fence} = Q3 + 1.5 \times \text{IQR}$$

Points beyond these fences are typically plotted as individual dots on a box plot.

OUTLIER DETECTION — WORKED EXAMPLE

Suppose $Q1 = 5$, $Q3 = 15 \rightarrow \text{IQR} = 15 - 5 = 10$

$$\text{Lower Fence} = 5 - 1.5 \times 10 = 5 - 15 = \mathbf{-10}$$

$$\text{Upper Fence} = 15 + 1.5 \times 10 = 15 + 15 = \mathbf{30}$$

Any value below -10 or above 30 would be flagged as an outlier.

PYTHON

```
df['col'].quantile(0.25) → Q1 · df['col'].quantile(0.75) → Q3 · sns.boxplot(x=df['col']) → draws the box plot (Seaborn)
```

DEFINITION

Skewness measures the **symmetry** (or lack of it) of a data distribution's shape. It tells us whether one tail of the distribution is longer or "fatter" than the other.

Skew Type	Shape	Mean vs Median	Skewness Value
Symmetric (zero skew)	Bell-shaped, both sides mirror each other	Mean $\approx$ Median $\approx$ Mode	≈ 0
Positive / Right Skew	Long tail stretches to the right	Mean $>$ Median	> 0
Negative / Left Skew	Long tail stretches to the left	Mean $<$ Median	< 0

Common Formula — Pearson's Coefficient of Skewness**FORMULA (EXTENDED THEORY)**

Pearson's first coefficient of skewness: $Sk = 3 \times (\text{Mean} - \text{Median}) / \text{Standard Deviation}$

The commonly-used "moment-based" skewness (what Pandas computes) is: $\text{Skewness} = [\Sigma(x_i - \bar{x})^3 / n] / s^3$

REAL-WORLD EXAMPLES

Positively skewed: Household income, house prices (a few very high earners/expensive homes pull the tail right)

Negatively skewed: Age at retirement, exam scores on an easy test (most cluster near the max, with a tail of low scorers pulling left)

Roughly symmetric: Adult human heights within a single gender

PYTHON (PANDAS)

```
df['column_name'].skew()
```

WHY SKEWNESS MATTERS FOR EDA

Many statistical techniques (like Pearson correlation and t-tests) assume roughly normal/symmetric data. Detecting skewness early tells you whether to use the **median** instead of the mean, apply a **transformation** (log, square-root) to reduce skew, or choose a non-parametric statistical test instead.

16

Kurtosis

Measuring the "tailedness" / "peakedness"

📘 DEFINITION

Kurtosis measures the **tailedness** / **peakedness** of a distribution compared to a normal distribution — specifically, how much of the data's variance comes from rare, extreme deviations (heavy tails) versus frequent, modest deviations.

Type	Peak Shape	Tails	Excess Kurtosis
Leptokurtic	Tall and sharp peak	Heavy / fat tails, more outliers	> 0
Mesokurtic	Medium peak (normal distribution)	Moderate tails — the baseline reference	= 0
Platykurtic	Flat, wide peak	Thin / light tails, fewer outliers	< 0

📐 FORMULA (EXTENDED THEORY)

$$\text{Kurtosis} = [\Sigma(x_i - \bar{x})^4 / n] / s^4$$

A perfectly normal distribution has kurtosis = 3. To make comparison to "normal" intuitive, most software (including Pandas) reports **Excess Kurtosis** = Kurtosis − 3, so a normal distribution shows as 0.

💡 MEMORY TRICK

Lepto (Greek: "thin/slender") → tall & narrow peak, fat tails. **Platy** (Greek: "flat/broad") → flat & wide peak, thin tails. **Meso** ("middle") → the balanced, normal case.

✔ PRACTICAL MEANING

In finance, a **leptokurtic** return distribution warns of "fat tail risk" — extreme market crashes/rallies happen more often than a normal distribution would predict, which is why relying purely on standard deviation can understate real-world risk.

🐍 PYTHON (PANDAS)

```
df['col'].kurt() — returns excess kurtosis by default
```

WHY VISUALIZE?

As statistician Francis Anscombe famously demonstrated with "Anscombe's Quartet" (4 datasets with nearly identical mean, variance, and correlation, but wildly different shapes when plotted), **summary statistics alone can hide critical patterns**. Visualization is often the fastest way to catch what numbers miss.

Chart	Best For	Variable Type(s)
Histogram	Distribution shape of a single numeric variable (frequency of value-ranges)	Univariate — Numerical
Box Plot	Spread, median, and outliers of a numeric variable (see Unit 14)	Univariate — Numerical
Bar Chart	Frequency/count comparison across categories	Univariate — Categorical
Pie Chart	Proportion of categories out of a whole (use sparingly — bar charts are usually clearer)	Univariate — Categorical
Scatter Plot	Relationship/correlation between two numeric variables (see Unit 19)	Bivariate — Numerical vs Numerical
Line Chart	Trend of a numeric variable over time	Bivariate — Time vs Numerical
Heatmap	Correlation matrix or 2D density across many variables at once	Multivariate
Pair Plot	Grid of scatter plots for every pair of numeric variables at once	Multivariate
Violin Plot	Combines a box plot with a density/distribution shape	Univariate/Bivariate

Histogram — Extended Theory

A histogram divides the range of a numeric variable into intervals ("bins") and shows the count (or frequency) of observations in each bin as bars. Unlike a bar chart, there are **no gaps** between histogram bars, because the x-axis is continuous.

CHOOSING THE NUMBER OF BINS

Too few bins hides detail; too many bins looks noisy. A common starting rule of thumb is **Sturges' Rule**: $k = 1 + \log_2(n)$, where n is the number of observations and k is the suggested bin count.

Bar Chart vs Histogram — Common Confusion

Feature	Bar Chart	Histogram
X-axis type	Categorical (discrete labels)	Numeric (continuous ranges/bins)
Bar spacing	Gaps between bars	No gaps (bars touch)
Order of bars	Can be rearranged freely	Fixed left-to-right numeric order

PYTHON (MATPLOTLIB / SEABORN)

```
df['col'].hist(bins=10) → histogram
sns.boxplot(x=df['col']) → box plot
df['category_col'].value_counts().plot(kind='bar') → bar chart
plt.scatter(df['x'], df['y']) → scatter plot
sns.heatmap(df.corr(), annot=True) → correlation heatmap
sns.pairplot(df) → pair plot
```

WHICH PLOT FOR WHICH QUESTION?

- "What does the distribution of Age look like?" → **Histogram** or **Box plot**
- "How many students are in each grade category?" → **Bar chart**
- "Is there a relationship between Study Hours and Score?" → **Scatter plot**
- "How does revenue change month by month?" → **Line chart**
- "Which pairs of variables in my dataset are most correlated?" → **Heatmap**

Types of Missing Data (extended theory)

Type	Meaning
MCAR (Missing Completely At Random)	Missingness has no pattern/relationship to any variable — pure chance
MAR (Missing At Random)	Missingness is related to another <i>observed</i> variable, but not the missing value itself
MNAR (Missing Not At Random)	Missingness is related to the missing value itself (e.g., high earners refusing to disclose income)

Common Ways to Handle Missing Data

Technique	Description	When to Use
Listwise Deletion	Drop any row containing a missing value	Missing data is a small % of the dataset and MCAR
Mean/Median Imputation	Replace missing numeric values with the column's mean or median	Simple, but reduces variance — median preferred for skewed data
Mode Imputation	Replace missing categorical values with the most frequent category	Categorical columns
Forward/Backward Fill	Carry the previous/next valid value forward or backward	Time-series data
Predictive Imputation	Use a model (regression/KNN) to estimate the missing value from other columns	When missingness relates to other known variables (MAR)

🔗 PYTHON (PANDAS)

```
df.isnull().sum() → count missing values per column
df.dropna() → listwise deletion
df['col'].fillna(df['col'].mean()) → mean imputation
df['col'].fillna(method='ffill') → forward fill
```

Outlier Detection — Recap & More Methods

Unit 14 introduced the **IQR rule** ($1.5 \times \text{IQR}$ fences) for outlier detection using box plots. Another widely-used statistical method is the **Z-score method**:

⚠️ Z-SCORE OUTLIER RULE

$$Z = (x_i - \bar{x}) / s$$

A common rule of thumb: flag any observation with $|Z| > 3$ as a potential outlier (i.e., more than 3 standard deviations away from the mean).

Method	Best For	Limitation
IQR / Box Plot Rule	Skewed data, doesn't assume normality	Can flag too many points in heavily skewed data
Z-score Rule	Roughly normal (symmetric) data	Less reliable when data is skewed or has extreme outliers already inflating the mean/SD

What To Do With Detected Outliers

- **Investigate first** — is it a genuine data-entry error (e.g., score = -20) or a real extreme value (e.g., a billionaire's income)?
- **Correct** — fix data-entry errors if the true value can be recovered
- **Remove** — drop the record if it's confirmed invalid/erroneous
- **Cap/Winsorize** — replace extreme values with a defined percentile boundary (e.g., cap at the 1st/99th percentile) rather than deleting the row
- **Keep & Transform** — if the value is genuine, consider a log/square-root transformation to reduce its influence rather than removing valuable data

⚠️ GOLDEN RULE OF OUTLIER TREATMENT

Never delete an outlier just because it's inconvenient. Always ask **why** it exists first — a true outlier can be the most informative data point in the whole dataset (e.g., detecting fraud, a rare disease case, or a breakthrough result).

These measures quantify the **relationship / association** between two variables, X and Y.

✓ LINEAR RELATIONSHIP

As X increases, Y changes at a **constant rate** — data points roughly follow a straight line on a scatter plot.

⚡ NON-LINEAR RELATIONSHIP

The relationship between X and Y follows a **curve**, not a straight line — a constant-rate assumption doesn't hold.

Covariance — The Building Block of Correlation

⚠ COVARIANCE FORMULA

$$\text{Cov}(X, Y) = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / (n - 1)$$

Covariance tells us the **direction** of a linear relationship (positive/negative), but its magnitude is hard to interpret because it depends on the units of X and Y. **Correlation solves this** by standardizing covariance into a unitless value between -1 and +1:

$$\text{Pearson's } r = \text{Cov}(X, Y) / (s_x \cdot s_y)$$

Scatter Plots — Reading Correlation Direction

Pattern	Meaning
Positive Correlation	Points trend upward (bottom-left to top-right) — as X increases, Y increases
Negative Correlation	Points trend downward (top-left to bottom-right) — as X increases, Y decreases
No / Non-linear Correlation	Points form a curve or random cloud — Pearson's r may be near zero even though a real (non-linear) pattern exists

Interpreting the Strength of Correlation (r)

Value of r	Strength
0.00 - 0.19	Very weak
0.20 - 0.39	Weak
0.40 - 0.59	Moderate
0.60 - 0.79	Strong
0.80 - 1.00	Very strong

🐍 PYTHON (PANDAS)

`df.corr()` — computes the **Pearson correlation coefficient** between all numeric column pairs by default.

⚠ CORRELATION ≠ CAUSATION

One of the most important principles in all of statistics: a strong correlation between X and Y does **not** prove that X causes Y. Classic example: ice-cream sales and drowning incidents are positively correlated — but ice cream doesn't cause drowning. Both are driven by a hidden **confounding variable**: hot summer weather.

⚠ IMPORTANT CAVEAT — NON-LINEARITY

Pearson correlation only captures **linear** relationships. A strong *non-linear* pattern between X and Y can still produce a Pearson r close to 0 — always look at the scatter plot, not just the number (this is the core lesson of Anscombe's Quartet from Unit 17).

Which Correlation Coefficient to Use?

The right correlation test depends on the **measurement scale** of the two variables being compared:

Variable X	Variable Y	Correct Test
Interval / Ratio	Interval / Ratio	Pearson's r
Category (Nominal / Ordinal)	Category (Nominal / Ordinal)	Spearman's r_s
Binomial (2-category)	Continuous	Point-Biserial Correlation
Binomial (2-category)	Binomial (2-category)	Cramér's φ (phi) / V

Spearman's Rank Correlation — Extended Theory

Spearman's r_s is essentially Pearson's r calculated on the **ranks** of the data rather than the raw values. This makes it robust to outliers and valid for

ordinal data or non-linear-but-monotonic relationships (where one variable consistently increases or decreases with the other, even if not at a constant rate).

PYTHON (SCIPY) IMPLEMENTATIONS

```
import scipy.stats as ss  
ss.pearsonr(x, y) → Pearson's r (with p-value)  
ss.spearmanr(x, y) → Spearman's rank correlation  
ss.pointbiserialr(x, y) → Point-biserial correlation  
pd.crosstab(x, y) → Contingency table, used as the basis for Cramér's  $\phi$  /  $V$  (also available via the statsmodels library)  
df.corr() → Pearson's r (default method in Pandas)
```

DEFINITION

The **Normal (Gaussian) Distribution** is a continuous, symmetric, bell-shaped probability distribution where the mean, median, and mode are all equal, and data clusters symmetrically around the center. Many real-world variables (height, measurement error, test scores) approximately follow this shape, which is why it's central to statistics.

The Empirical Rule (68-95-99.7 Rule)

Range Around Mean	% of Data Contained
$\bar{x} \pm 1 \text{ SD}$	$\approx 68\%$
$\bar{x} \pm 2 \text{ SD}$	$\approx 95\%$
$\bar{x} \pm 3 \text{ SD}$	$\approx 99.7\%$

This rule is precisely why the Z-score outlier rule from Unit 18 uses a threshold of $|Z| > 3$ — beyond 3 standard deviations, only about 0.3% of data should naturally occur in a normal distribution, making such points highly suspicious as outliers.

The Central Limit Theorem (CLT) — Why Normal Distribution Is So Important**CORE THEORY**

The **Central Limit Theorem** states that if you repeatedly take sufficiently large random samples from *any* population (regardless of its original distribution shape) and calculate their means, the distribution of those sample means will approximate a **normal distribution**. This is the theoretical foundation that allows inferential statistics (confidence intervals, hypothesis tests) to work even when the underlying population isn't normal.

Connecting Normal Distribution to Earlier Units

- **Skewness ≈ 0** and **Excess Kurtosis ≈ 0** together are common (though not sufficient on their own) indicators that data may be approximately normal (Units 15-16)
- Many statistical tests (Pearson correlation, t-tests) formally **assume normality** — this is why checking skewness/kurtosis/histograms during EDA matters before applying them
- The Z-score method for outlier detection (Unit 18) is built directly on normal-distribution theory

PYTHON — QUICK NORMALITY CHECK

```
df['col'].hist() → visually inspect shape
scipy.stats.shapiro(df['col']) → Shapiro-Wilk normality test (small/medium samples)
scipy.stats.normaltest(df['col']) → D'Agostino's K2 normality test
```

Measurement Scales (NOIR)

Scale	Order?	Diff?	True Zero?	Valid Center	Example
Nominal	✗	✗	✗	Mode	Gender, Blood Group, City
Ordinal	✓	✗	✗	Median	Satisfaction, Rank
Interval	✓	✓	✗	Mean	Temperature (°C/°F)
Ratio	✓	✓	✓	Mean	Age, Height, Weight, Salary

Key Formulas

Measure	Formula
Mean	$\bar{x} = \sum x_i / n$
Median (n odd / even)	$x_{((n+1)/2)} / \frac{1}{2}[x_{(n/2)} + x_{(n/2+1)}]$
Weighted Mean	$\sum(w_i x_i) / \sum w_i$
Range	$X_H - X_L$
Population Variance	$\sigma^2 = \sum(x_i - \bar{x})^2 / N$
Sample Variance	$s^2 = \sum(x_i - \bar{x})^2 / (n - 1)$
Standard Deviation	$\sqrt{(\text{Variance})}$
Coefficient of Variation	$(s / \bar{x}) \times 100\%$
Percentile	$P(x_i) = [(i - 1) / (n - 1)] \times 100$
IQR	$Q3 - Q1$
Quartile Deviation	$(Q3 - Q1) / 2$
Outlier Fences (IQR rule)	$Q1 - 1.5 \cdot IQR / Q3 + 1.5 \cdot IQR$
Z-score	$(x_i - \bar{x}) / s$
Covariance	$\sum[(x_i - \bar{x})(y_i - \bar{y})] / (n - 1)$
Pearson's r	$\text{Cov}(X, Y) / (s_x \cdot s_y)$
Skewness (Pearson's 1st coeff.)	$3(\text{Mean} - \text{Median}) / \text{SD}$

Correlation Chooser

X type	Y type	Use
Interval/Ratio	Interval/Ratio	Pearson r
Ordinal/Nominal	Ordinal/Nominal	Spearman r _s
Binomial	Continuous	Point-Biserial
Binomial	Binomial	Cramér's ϕ/V

Distribution Shape Quick-Reference

Skewness	Kurtosis	Shape
≈ 0	≈ 0 (excess)	Normal / Mesokurtic
> 0	—	Right/Positive skew — Mean $>$ Median
< 0	—	Left/Negative skew — Mean $<$ Median
—	> 0	Leptokurtic — sharp peak, fat tails
—	< 0	Platykurtic — flat peak, thin tails

Python One-Liners

Task	Code
Load data	<code>pd.read_csv("file.csv")</code>
Dataset overview	<code>df.info() / df.describe()</code>

Missing values	<code>df.isnull().sum()</code>
Mean / Median / Mode	<code>df['c'].mean()/median()/mode()</code>
Std Dev / Variance	<code>df['c'].std() / .var()</code>
Skewness / Kurtosis	<code>df['c'].skew() / .kurt()</code>
Quartiles	<code>df['c'].quantile(0.25)</code>
Histogram / Box plot	<code>df['c'].hist() / sns.boxplot()</code>
Pearson correlation matrix	<code>df.corr()</code>
Spearman correlation	<code>scipy.stats.spearmanr(x, y)</code>
Point-biserial correlation	<code>scipy.stats.pointbiserialr(x, y)</code>
Contingency table	<code>pd.crosstab(x, y)</code>

Section A — Short / Conceptual Questions

Q1. Differentiate between Data and Information.

Data is raw, unprocessed facts (e.g., 17, 41, 30, 99) that carry no inherent meaning. Information is the meaningful output obtained after processing and analyzing that data (e.g., "scores with an average of 50"). Data and information are complementary — data provides raw facts, information gives them meaning. This maps onto the broader DIKW hierarchy (Data → Information → Knowledge → Wisdom).

Q2. Explain Primary, Secondary and Tertiary data with examples.

Primary data is collected first-hand by the researcher (e.g., conducting your own survey) — "I am involved." **Secondary data** is collected by someone else but reused by you (e.g., a government census report) — "You are involved." **Tertiary data** is a compilation of primary and secondary data (e.g., an encyclopedia or literature review).

Q3. What is the difference between IDA, EDA and CDA?

IDA covers problem definition, data collection and preprocessing/cleaning to make data usable. **EDA** examines statistical aspects (mean, variance, mode, distribution) and processes data further (cleaning, missing values, outliers, transformation), identifying variable features. **CDA** tests hypotheses and statistically confirms relationships between variables, including modelling and evaluation.

Q4. Why is temperature (in °C) an Interval scale and not a Ratio scale?

Temperature in Celsius has meaningful, equal differences, but lacks a true/absolute zero — 0°C does not mean "no temperature," it's an arbitrary reference point. Therefore, ratio statements like "20°C is twice as hot as 10°C" are invalid, which is why it's Interval, not Ratio.

Q5. Why does Age qualify as a Ratio scale variable?

Age has an absolute zero (age = 0 at birth genuinely means "no time has elapsed"), so meaningful ratio comparisons are valid — a 40-year-old is genuinely twice as old as a 20-year-old.

Q6. What is a Likert scale, and what measurement scale does it belong to?

A Likert scale (e.g., Strongly Disagree → Strongly Agree, typically 3/5/7-point) is fundamentally an **Ordinal** scale. It is often transformed to be treated as an Interval scale by assigning numbers to each option and treating the gaps between them as equal in length.

Q7. Why do we divide by (n – 1) instead of n when calculating sample variance?

This is called Bessel's Correction. A sample tends to underestimate the true spread of the full population, since sample data points are, on average, closer to the sample mean than to the true population mean. Dividing by (n – 1) instead of n compensates for this bias.

Q8. Distinguish between Skewness and Kurtosis.

Skewness measures the asymmetry of a distribution — whether it's stretched more to the left or right. **Kurtosis** measures the "tailedness"/"peakedness" — how sharp the peak is and how heavy the tails are, compared to a normal distribution.

Q9. Why can't Pearson's correlation coefficient always detect a real relationship?

Pearson's *r* only measures the strength of a **linear** relationship. A strong **non-linear** relationship between X and Y can produce a Pearson *r* close to zero even though a clear pattern exists — always visualize with a scatter plot too.

Q10. Which correlation coefficient would you use between "Education Level" (ordinal) and "Job Satisfaction" (ordinal)?

Since both variables are categorical/ordinal, **Spearman's rank correlation (*r_s*)** is the appropriate choice, not Pearson's *r*.

Q11. What is the difference between a population and a sample? Why does it matter for EDA?

A population is the entire group of interest; a sample is a subset actually observed. It matters because formulas differ (population variance divides by *N*, sample variance by *n*–1), and because inferential statistics use the sample to estimate unknown population parameters.

Q12. What is the "Garbage In, Garbage Out" principle?

It states that flawed, poor-quality input data will always produce untrustworthy output, no matter how sophisticated the analysis or model. This is why data validation (validity, reliability, accuracy) during IDA is essential before any further analysis.

Q13. Explain the difference between MCAR, MAR, and MNAR missing data.

MCAR — missingness is pure chance, unrelated to any variable. **MAR** — missingness relates to another observed variable, not the missing value itself. **MNAR** — missingness relates to the missing value itself (e.g., high earners not disclosing income).

Q14. What does the Central Limit Theorem state, and why is it important?

It states that the distribution of sample means approaches a normal distribution as sample size increases, regardless of the population's original distribution shape. This underpins most inferential statistics techniques (confidence intervals, hypothesis tests) even when the population isn't normally distributed.

Q15. Why is "correlation does not imply causation" important in EDA?

A strong statistical relationship between two variables doesn't prove one causes the other — a hidden confounding variable may drive both (e.g., ice-cream sales and drowning incidents are correlated due to hot weather, not because one causes the other). EDA can reveal associations but cannot, by itself, establish causation — that typically requires controlled experiments.

Section B — Multiple Choice Questions (MCQs)

1. Which of the following is an example of Nominal data?

- A. Exam rank B. Temperature **C. Blood group** ✓ D. Height

2. Which measure of central tendency does NOT require sorting the data?

- A. Mean** ✓ B. Median C. Mode D. All require sorting

3. A distribution with a long tail to the right is called:

- A. Negatively skewed **B. Positively skewed** ✓ C. Symmetric D. Platykurtic

4. A sharply-peaked distribution with heavy tails is called:

- A. Leptokurtic** ✓ B. Mesokurtic C. Platykurtic D. Skewed

5. Which measurement scale has a true/absolute zero?

- A. Nominal B. Ordinal C. Interval **D. Ratio** ✓

6. In a structured dataset, what do rows represent?

- A. Variables **B. Observations** ✓ C. Features D. File format

7. Which Pandas function computes the Pearson correlation matrix of a DataFrame?

- A. df.mean() B. df.skew() **C. df.corr()** ✓ D. df.describe()

8. Which file format stores data as key-value pairs?

- A. CSV B. XLSX **C. JSON** ✓ D. TXT

9. For two binomial (2-category) variables, which correlation measure is appropriate?

- A. Pearson r B. Point-biserial C. Spearman r_s **D. Cramér's ϕ** ✓

10. Which stage of analysis focuses on testing hypotheses and confirming relationships?

- A. IDA B. EDA **C. CDA** ✓ D. None of these

11. According to the Empirical Rule, approximately what % of normally distributed data falls within ± 2 SD of the mean?

- A. 68% **B. 95%** ✓ C. 99.7% D. 50%

12. Which chart is best suited to show the distribution shape of a single continuous numeric variable?

- A. Pie chart **B. Histogram** ✓ C. Bar chart D. Contingency table

13. In the IQR outlier rule, what is the Upper Fence formula?

- A. $Q1 - 1.5 \times IQR$ **B. $Q3 + 1.5 \times IQR$** ✓ C. $Mean + 3 \times SD$ D. $Median + IQR$

14. Who is credited with formally coining "Exploratory Data Analysis" and inventing the box plot?

- A. Karl Pearson B. Rensis Likert **C. John W. Tukey** ✓ D. Ronald Fisher

15. A strong correlation between two variables proves that one causes the other. True or False?

- A. True **B. False** ✓ — correlation does not imply causation; a confounding variable may explain both

Section C — Numerical Problems

Problem 1: Given the dataset 4, 8, 6, 5, 3, 7, compute the Mean, and identify the Median.

Mean = $(4+8+6+5+3+7)/6 = 33/6 = 5.5$

Sorted data: 3, 4, 5, 6, 7, 8 ($n = 6$, even)

Median = $\frac{1}{2}(x_3 + x_4) = \frac{1}{2}(5 + 6) = 5.5$

Problem 2: For the dataset 2, 3, 5, 6, 8, 10, 12, 15, 18, 20 ($n = 10$), find the percentile rank of the value 6.

Position of 6 in sorted data: $i = 4$.

$P(x_4) = [(4 - 1)/(10 - 1)] \times 100 = (3/9) \times 100 = 33.3\%$

Problem 3: For the dataset 2, 3, 5, 6, 8, 10, 12, 15, 18, 20 ($n = 10$), find the 75th percentile position (i).

$i - 1 = [P(x_i) \times (n - 1)] / 100 = (75 \times 9)/100 = 675/100 = 6.75$

$i = 7.75$ (lies 75% of the way between the 7th value (12) and 8th value (15))

Problem 4: A population dataset has values 5, 7, 9, and mean = 7. Compute the population variance and standard deviation.

Deviations: $(5-7)^2 = 4$, $(7-7)^2 = 0$, $(9-7)^2 = 4 \rightarrow \text{Sum} = 8$

Population Variance $\sigma^2 = 8 / 3 = 2.67$

Standard Deviation $\sigma = \sqrt{2.67} \approx 1.63$

Problem 5: Using the same values (5, 7, 9) as a SAMPLE instead of a population, compute the sample variance.

Sum of squared deviations = 8 (same as above)

Sample Variance $s^2 = 8 / (3 - 1) = 8/2 = 4.0$ (larger than the population variance — the effect of Bessel's correction)

Problem 6: For the dataset with $Q1 = 20$ and $Q3 = 50$, compute the IQR, Quartile Deviation, and the outlier fences.

IQR = $Q3 - Q1 = 50 - 20 = 30$

Quartile Deviation = $\text{IQR}/2 = 30/2 = 15$

Lower Fence = $Q1 - 1.5 \times \text{IQR} = 20 - 45 = -25$

Upper Fence = $Q3 + 1.5 \times \text{IQR} = 50 + 45 = 95$

Problem 7: A data point has value 90, the dataset mean is 60, and the standard deviation is 10. Compute its Z-score and state whether it's a likely outlier.

$Z = (90 - 60)/10 = 3.0$

Since $|Z| = 3$ is at the common threshold ($|Z| > 3$ flags an outlier), this point is a **borderline/likely outlier** and should be investigated further.

Problem 8: Two employees' salaries have Mean = ₹50,000 and Median = ₹42,000. What does this indicate about the distribution's skewness?

Since Mean > Median, the distribution is **positively (right) skewed** — a few very high salaries are pulling the mean upward, while most employees earn closer to the (lower) median.

📌 FINAL REVISION CHECKLIST

- ✓ Can you explain Data vs Information (and the DIKW hierarchy) with an example?
- ✓ Can you list Primary / Secondary / Tertiary data with the involvement rule?
- ✓ Can you explain IDA → EDA → CDA in one line each, and Tukey's EDA vs CDA distinction?
- ✓ Can you classify any variable into Nominal / Ordinal / Interval / Ratio using the 4-step test?
- ✓ Can you write the Mean, Median, Variance, SD, CV, Percentile, IQR, Z-score and Correlation formulas from memory?
- ✓ Can you pick the correct correlation coefficient for any pair of variable types?
- ✓ Can you explain the difference between skewness and kurtosis, with real examples?
- ✓ Can you list at least 3 techniques for handling missing data and 2 methods for detecting outliers?
- ✓ Can you explain the Empirical Rule and the Central Limit Theorem in your own words?
- ✓ Can you recall the relevant Python one-liners (`df.corr()`, `df['col'].skew()`, `scipy.stats` functions)?

📖 FOUNDATIONAL TEXTS & PAPERS

- Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley.
- Stevens, S. S. (1946). "On the Theory of Scales of Measurement." *Science*, 103(2684), 677-680.
- Likert, R. (1932). "A Technique for the Measurement of Attitudes." *Archives of Psychology*, 22(140), 1-55.
- Anscombe, F. J. (1973). "Graphs in Statistical Analysis." *The American Statistician*, 27(1), 17-21. (Source of "Anscombe's Quartet")
- Wickham, H. & Grolemund, G. *R for Data Science* — widely-referenced open text on the data science / EDA workflow (freely available online).
- McKinney, W. *Python for Data Analysis* — the standard reference for the Pandas library, written by its creator.

🌐 USEFUL ONLINE REFERENCES & DOCUMENTATION

- Pandas official documentation — pandas.pydata.org
- SciPy Statistics reference — docs.scipy.org/doc/scipy/reference/stats.html
- Seaborn statistical visualization docs — seaborn.pydata.org
- Kaggle "Learn" micro-courses on Pandas & Data Visualization — free, hands-on practice
- UCI Machine Learning Repository & Kaggle Datasets — for practicing EDA on real, open datasets
- CRISP-DM methodology overview — the industry-standard data mining process framework

🔗 HOW TO USE THESE REFERENCES

These sources are provided so you can go deeper into any topic beyond what's required for the exam — especially useful if you want to build real EDA projects using the Python tools introduced throughout this note (Pandas, NumPy, Matplotlib, Seaborn, SciPy).